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Experiment (4) Report

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Faculty of Engineering - Ain Shams University
Electronics and Electrical Communications Engineering Department

ECE352s Analog and Digital Communication Systems

Junior - Electronics and Electrical Communications

Experiment (4)

Digital Passband Data Transmission: BASK, BFSK, BPSK

Spring 2026

1. Objectives:

- Model Digital Data by Sequence Generator.
- Transmit data by different modulation schemes.
- Recover the data using coherent and non-coherent detection techniques.

2. Equipment:

- Digital CRO
- Telecoms-Trainer 101 KIT

3. Part 1: Binary Amplitude Shift Keying

Sharing the channel transmission bandwidth is essential and there are several methods of doing so.

One is called time division multiplexing (TDM) and involves giving the users exclusive access to the channel for short periods of time. On the face of it, this type of sharing might seem impractical. Imagine giving all mobile phone users in a cell just a minute or so to make their call then having to wait until their turn comes around again. However, TDM works well when the access time is extremely short (less than a second) and the rate of the sharing is fast. This allows multiple users to appear to have access all at the same time. TDM is used for digital communications and is achieved by interleaving the users' data. That is, a portion of one user's data is transmitted followed by a portion of the next user's data and so on. Unfortunately, there's a catch. If the message is real-time information which can't afford to be delayed (like digitally encoded speech) then, as the number of users increases, so must the data's bit-rate.

Another method of sharing the channel is called frequency division multiplexing (FDM) and involves giving the users exclusive and uninterrupted access to a portion of the channel's radio frequency spectrum. To transmit their message the user must superimpose it onto a carrier that sits inside their allocated band of frequencies. This method is used by broadcast radio and television to share free-space. FDM is also used for digital communications and uses the same modulation schemes available to analog communications including: AM, DSBSC and FM. When AM is used for multiplexing digital data, it is known as amplitude shift keying (ASK). Other names include: on-off keying, continuous wave and interrupted continuous wave.

Figure 1 below shows what an ASK signal looks like time-coincident with the digital signal that has been used to generate it.

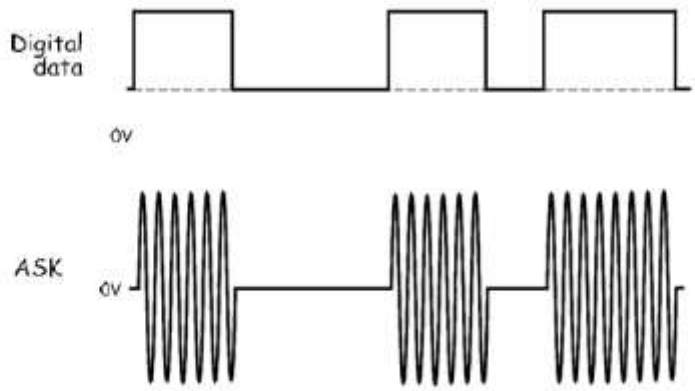


Figure 1

Notice that the ASK signal's upper and lower limits (the envelopes) are the same shape as the data stream (though the lower envelope is inverted). This is simultaneously an advantage and a disadvantage of ASK. Recovery of the data stream can be implemented using a simple envelope detector, However, noise on the channel can change the envelopes' shape enough for the receiver to interpret the logic levels incorrectly causing errors (analog AM communications have the same problem.

ASK can be generated using the digital signal to switch the carrier's connection to the channel on and off.

In this experiment you'll use the Emona Telecoms-Trainer 101 to generate an ASK signal using the switching method. Digital data for the message is modelled by the Sequence Generator module. You'll then recover the data using a simple envelope detector and observe its distortion. Finally, you'll use a comparator to restore the data.

Procedure

Part A - Generating an ASK signal

1. Set the scope's Channel 1 and Channel 2 Input Coupling controls to the DC position.
2. Locate the VCO module and set its Frequency Adjust control to about the middle of its travel.
3. Set the VCO module's Range control to the HI position.
4. Connect the set-up shown in Figure 2 below (Dotted lines is leads already in place)

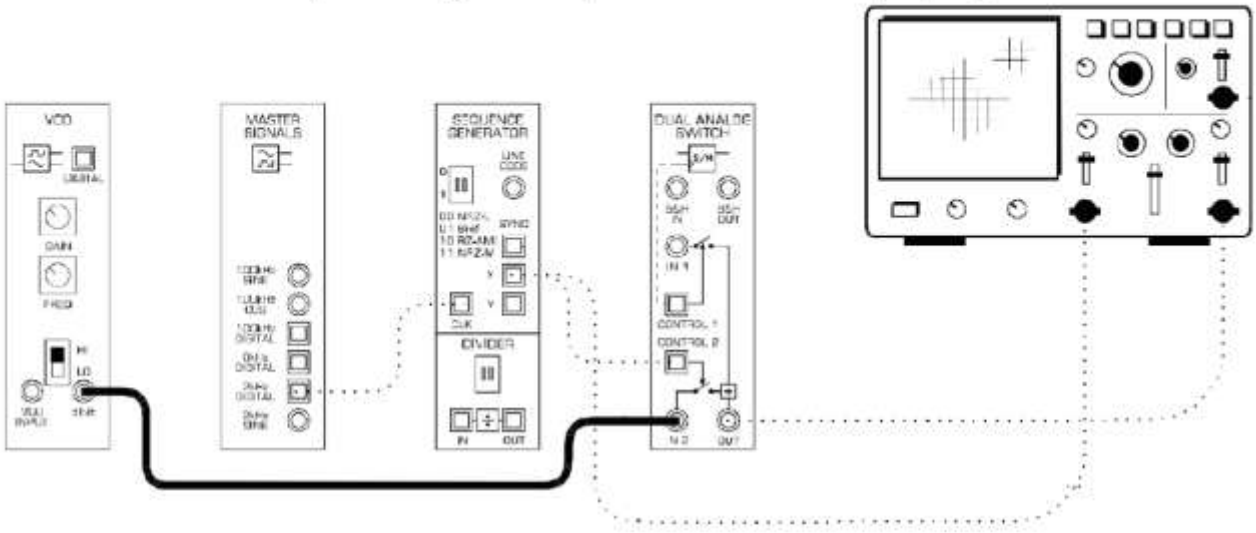


Figure 2

5. Use the scope's Channel 1 Vertical Position control to overlay the sequence generator module's output and the ASK signal out of the dual analog switch module. Comment.

What feature of the ASK signal suggests that it's an AM signal?

The **envelope** of the ASK signal (the shape formed by the peaks of the carrier) perfectly matches the shape of the digital message signal. Because the amplitude of the carrier is being varied in response to the message, it is functionally a form of AM where the "depth" is 100\% (on/off).

Part B - Demodulating an ASK signal using an envelope detector and a comparator

As ASK is really just AM (with a digital message instead of speech or music), it can be recovered using any of AM demodulation schemes.

1. Locate the Tuneable Low-pass Filter module and turn its Gain and cut-off frequency control fully clockwise.
2. Modify the set-up as shown in Figure 3 below.
   Note: the left most modules have been left off to fit the drawing in the page.
3. Set the variable VDC adjust control to about middle of its travel.
4. Compare the generated Sequence and the recovered output signal.
   if they are not the same, vary the variable VDC control until they are.

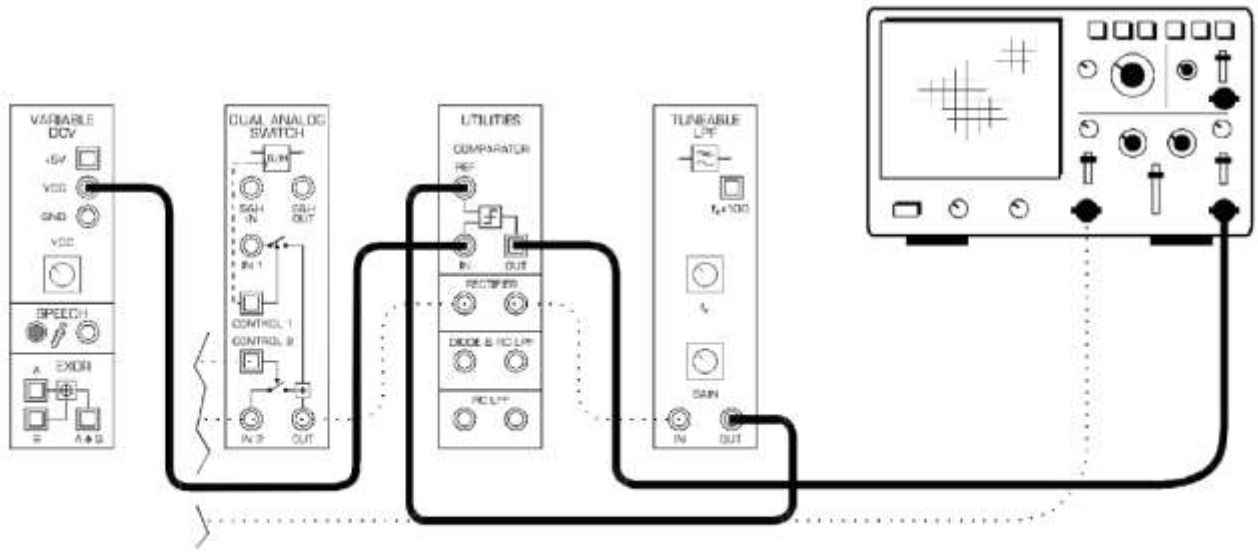


Figure 3

Why the output signal at the tuneable filter module is not a perfect copy of the original signal? What did the comparator add to enhance the output signal?

The **envelope** of the ASK signal (the shape formed by the peaks of the carrier) perfectly matches the shape of the digital message signal. Because the amplitude of the carrier is being varied in response to the message, it is functionally a form of AM where the "depth" is 100\% (on/off).

The **Comparator** performs a "hard decision" by comparing the rounded, noisy recovered signal against a reference DC voltage. It ignores the rounding and noise (as long as it stays below the threshold) and outputs a clean, sharp square wave with vertical edges, effectively **regenerating** the original digital data.

signal + noise), vary the amount of noise power added to signal and compare the recovered and original data in the presence of channel noise.

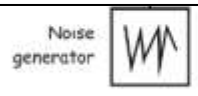


Figure 4

Is it still possible to recover the data at high channel noise?

Up to a certain point, yes. However, if the noise power is high enough that the noise peaks cross the comparator's threshold during a "Logic 0" (or dip below it during a "Logic 1"), the comparator will trigger incorrectly, causing **bit errors**

4. Part 2: Binary Frequency Shift Keying

Frequency division multiplexing (FDM) allows a channel to be shared among a set of users.

Recall that this is achieved by superimposing the message onto a carrier signal inside the user's allocated portion of the radio-frequency spectrum. Recall also that any of the analog modulation schemes can be used to transmit digital data in this way. When frequency modulation (FM) is used it is known as binary frequency shift keying (BFSK or more commonly just FSK). One of the reasons for using FSK is to take advantage of the relative noise immunity that FM enjoys over AM.

Recall that noise manifests itself as variations in the transmitted signal's amplitude. These variations can be removed by FM/FSK receivers (**by a circuit called a limiter**) without adversely affecting the recovered message. Figure 5 below shows what an FSK signal looks like time-coincident with the digital signal that has been used to generate it. Notice that the FSK signal switches between two frequencies. The frequency of the signal that corresponds with logic 0 is in the digital data (called the space frequency) is usually lower than the modulator's nominal carrier frequency. The frequency of the signal that corresponds with logic 1 is in the digital data (called the mark frequency) is usually higher than the modulator's nominal carrier frequency.

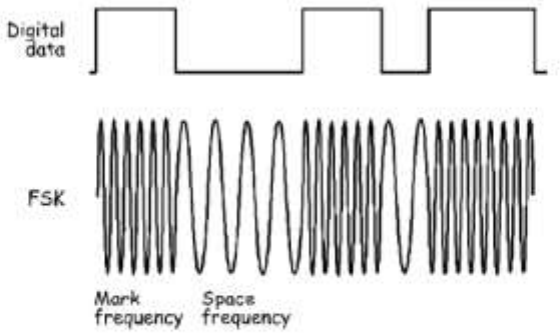


Figure 5

The modulator doesn't output a signal at the carrier frequency hence the reference here to it as being the "nominal" carrier frequency. FSK generation can be handled by conventional FM modulator circuits and the voltage- controlled oscillator (VCO) is commonly used. Similarly, FSK demodulation can be handled by conventional FM demodulators such as the zero-crossing detector and the phase-locked loop. Alternatively, if the FSK signal is passed through a sufficiently selective filter, the two sinewaves that make it up can be individually picked out. Considered on their own, each signal is an ASK signal and so the data can be recovered by passing either one of them through an envelope detector.

In this experiment you'll use the Emona Telecoms-Trainer 101 to implement the VCO method of generating an FSK signal. Digital data for the message is modelled by the Sequence Generator module. You'll then recover the data by using a filter to pick-out one of the sinewaves in the FSK signal and demodulate it using an envelope detector. Finally, you'll observe the demodulated FSK signal's distortion and use a comparator to restore the data.

Part A – Generating an FSK Signal

1. Set the scope's Channel 1 and Channel 2 Input Coupling controls to the DC position.
2. Locate the VCO module and set its Gain control to about half its travel.
3. Set the VCO module's Frequency Adjust control to about a quarter of its travel.
4. Set the VCO module's Range control to the LO position.
5. Locate the Sequence Generator module and set its dip-switches to 00.
   Tip: To do this, push both switches up.
6. Connect the setup shown in figure 6.

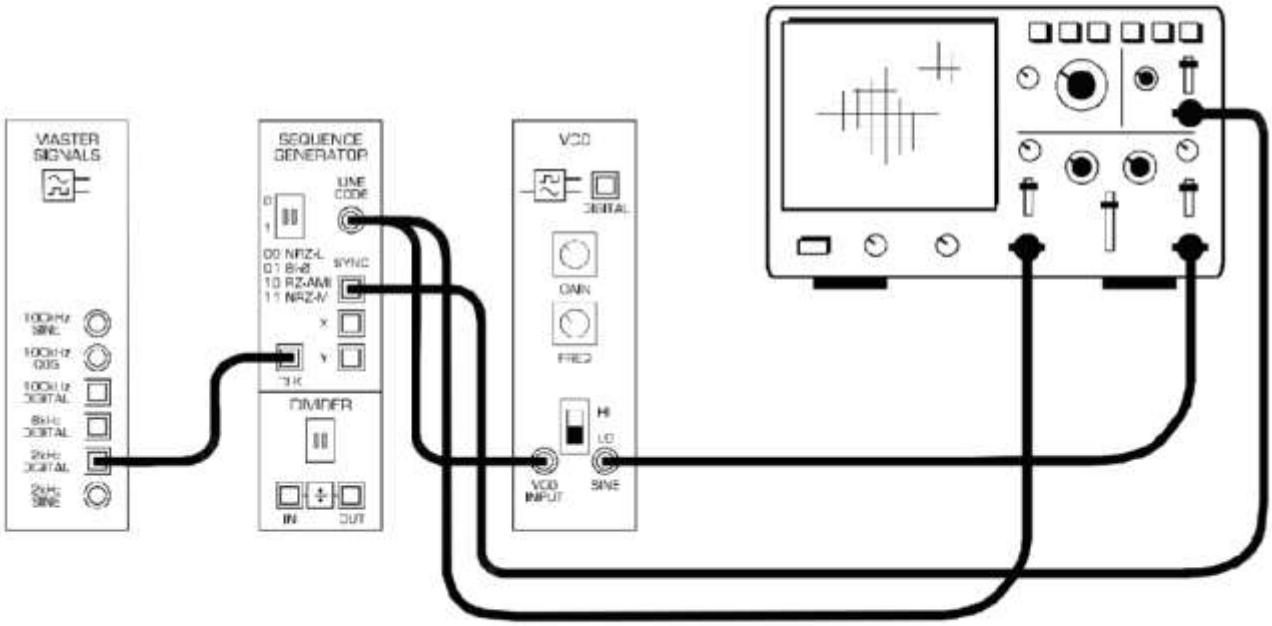


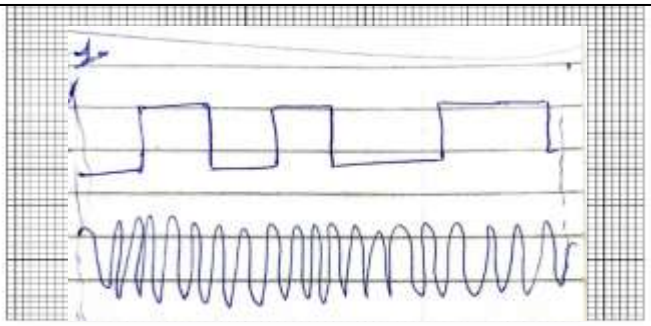
Figure 6

7. Use the scope's CH1 and CH2 to view and compare the sequence generator and the VCO output signals.

What's the name for the VCO output frequency that corresponds with logic 1 and logic 0 in the digital data?

Logic 1 is called the **Mark frequency**, and Logic 0 is called the **Space frequency**.

8. Draw the sequence generated signal and the FSK signal.



Part B - Demodulating an FSK signal using filtering and an envelope detector

As FSK is really just FM (with a digital message instead of speech or music), it can be recovered using any of the FM demodulation schemes. However, as the FSK signal switches back and forth between just two frequencies we can use a method of demodulating it that cannot be used to demodulate speech encoded FM signals.

1. Turn the VCO module's Frequency Adjust control to about the position of the number 2 on a clock face.
2. Locate the Tuneable LPF module and turn its Cut-off Frequency Adjust control fully clockwise.
3. Turn the Tuneable LPF's Gain control fully clockwise.
4. Modify the set-up as shown in Figure 7.
   Note: Remember that the dotted lines show leads already in place. Also, the left most modules have been cut-off to fit the drawing on the page.

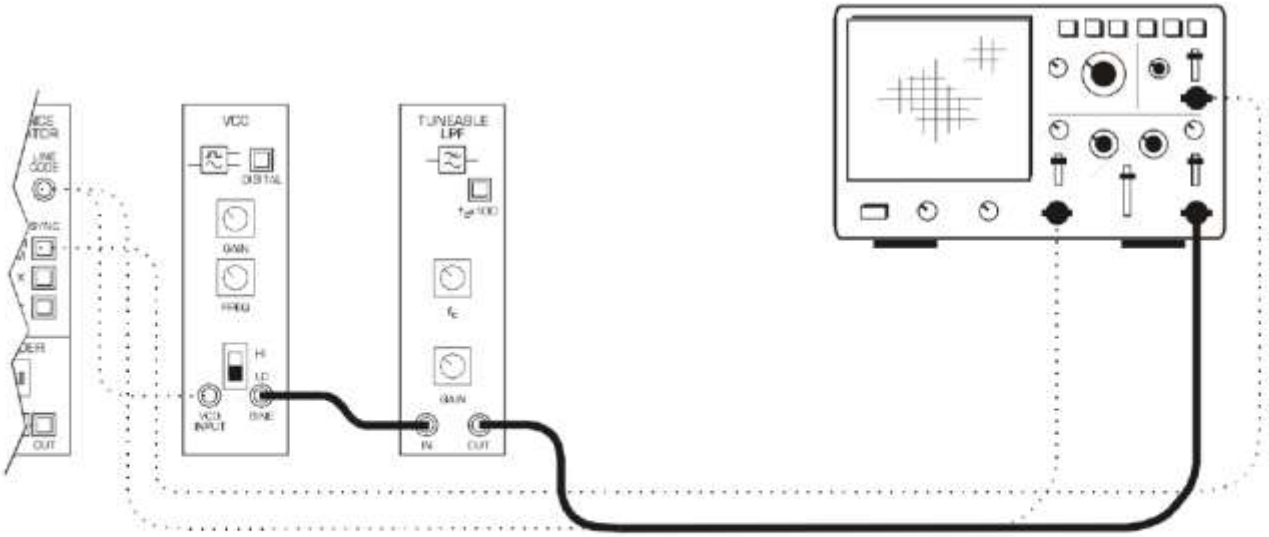


Figure 7

5. Turn the Tuneable LPF's Cut-off Frequency Adjust control slowly from fully clockwise until the higher frequency component of the FSK signal becomes equal to zero, as shown in Figure 8.
6. Compare the digital signal and the filter's output. These should be visible as Ch.1 and Ch.2 on the scope display.

Which of the FSK signal's two sinewaves is the filter picking out?

By slowly turning the **Tuneable LPF** cut-off frequency control anti-clockwise, the filter begins to reject the higher frequency (Mark) while still passing the lower frequency (Space). Thus, it is picking out the **Space frequency**.

What does the filtered FSK signal look like?

It looks like an ASK signal. You will see a sinewave during the Logic 0 periods of the message and nearly zero volts during the Logic 1 periods (where the Mark frequency was filtered out).

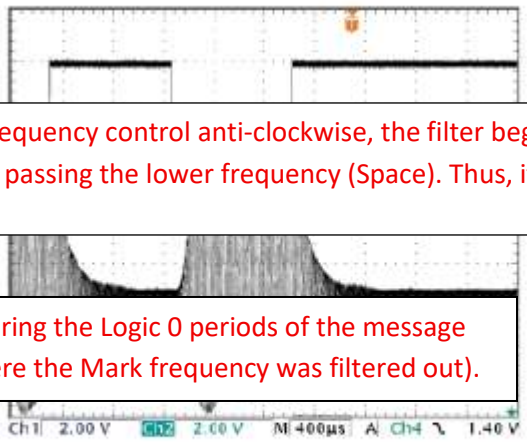


Figure 8

Envelope Detection:

The envelope detector used in this experiment differs from the simpler DIODE & RC LPF used in the earlier AM experiment. In this experiment we use a combination of the RECTIFIER followed by the BASEBAND LPF. As FSK uses digital data, a filter which is more responsive to the digital data stream is needed.

7. Modify the setup as shown in Figure 9 to include the envelope detector and comparator.

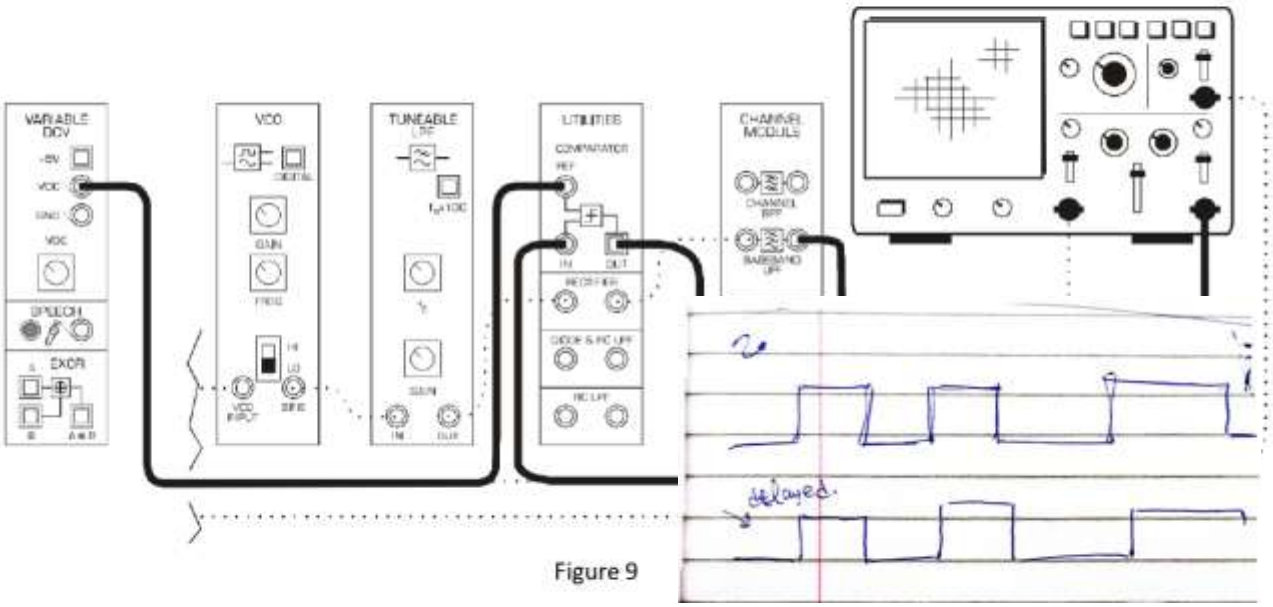


Figure 9

8. Compare the original digital signal and the recovered digital signal out of the comparator. If they're not the same vary the Variable VDC module's Variable DC control slightly until they are.

How does the comparator turn the slow rising voltages of the recovered digital signal into sharp transitions?

The comparator achieves sharp transitions by continuously comparing the sluggish, RC-filtered input voltage against a fixed DC reference voltage set by the user. The moment the rising input voltage crosses above this specific threshold, the comparator's internal circuitry immediately snaps its output to a predefined "High" logic level, and conversely snaps it to a "Low" level the instant the voltage falls below the threshold. This binary threshold-triggering mechanism completely ignores the gradual analog charging and discharging slopes of the envelope detector, allowing it to reconstruct the clean, vertical edges required for a perfect digital square wave.

9. Can you Introduce noise to the FSK signal, and recover it? Modify your setup to include channel noise, hence compare the original data with recovered signal, and comment on results.

Yes, you can successfully recover the FSK data even after introducing significant channel noise because FSK encodes its digital information in the frequency of the carrier, not its amplitude. When you add noise via the Adder module, the oscilloscope will display a highly distorted, messy FSK waveform with fluctuating peaks, but the combination of the frequency-selective Tuneable LPF and the comparator strips this amplitude noise away. Because the comparator makes a hard binary decision based only on whether the filtered envelope crosses your set voltage threshold, the final recovered output will remain a clean, highly accurate copy of the original digital sequence, clearly demonstrating FSK's

5. Part 3: Binary Phase Shift Keying

An alternative to BASK and BFSK two methods is to use the data stream's 1s and 0s to switch the carrier between two phases. This is called Binary Phase Shift Keying (BPSK). Figure 10 below shows what a BPSK signal looks like time-coincident with the digital signal that has been used to generate it. Notice that, when the change in logic level causes the BPSK signal's phase to change, it does so by 180°. For example, where the signal is travelling towards a positive peak the change in logic level causes it to reverse direction and head back toward the negative peak (and vice versa).

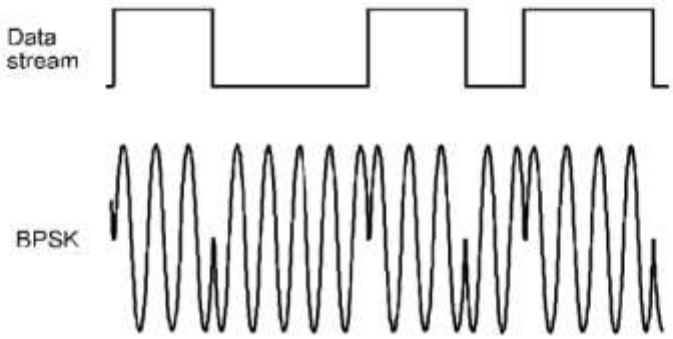


Figure 10

You may find it difficult to see at first but look closely and you'll notice that alternating halves of the BPSK signal's envelopes have the same shape as the message. This indicates that BPSK is actually double-sideband suppressed carrier (DSBSC) modulation. That being the case, BPSK generation and the recovery of the data can be handled by conventional DSBSC modulation and demodulation techniques.

With a choice of ASK, FSK and BPSK you might be wondering about which system you'll most likely see. All other things being equal, BPSK is the best performing system in terms of its ability to ignore noise and so it produces the fewest errors at the receiver. FM is the next best and AM is the worst. On that basis, you'd expect that BPSK is the preferred system.

However, it's not necessarily the easiest to implement and so in some situations FSK or ASK might be used as they are cheaper to implement. In fact, FSK was used for cheaper dial-up modems.

In this experiment you'll use the Emona Telecoms-Trainer 101 to generate a BPSK signal using the Multiplier module to implement its mathematical model. Digital data for the message is modelled by the Sequence Generator module. You'll then recover the data using another Multiplier module and observe its distortion. Finally, you'll use a comparator to restore the data.

Part A – Generating a BPSK Signal

A BPSK signal will be generated by implementing its mathematical model.

1. Set the scope's Channel 1 and Channel 2 Input Coupling controls to the DC position.
2. Locate the Sequence Generator module and set its dip-switches to 00.
3. Connect the set-up shown in Figure 11.

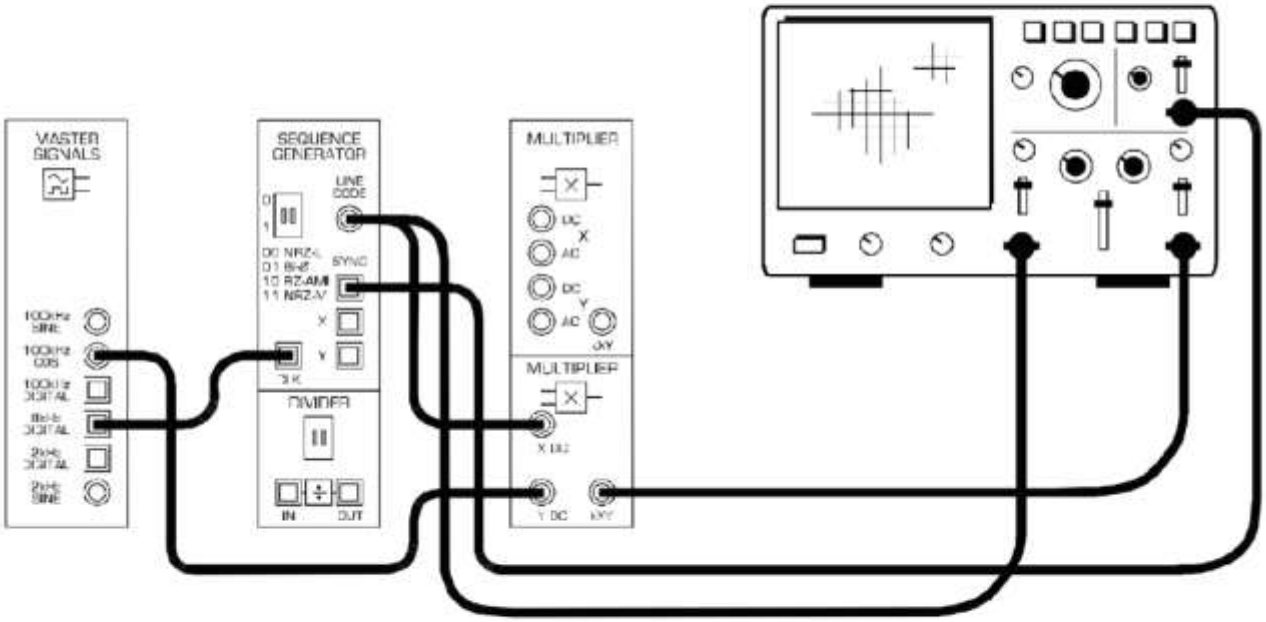


Figure 11

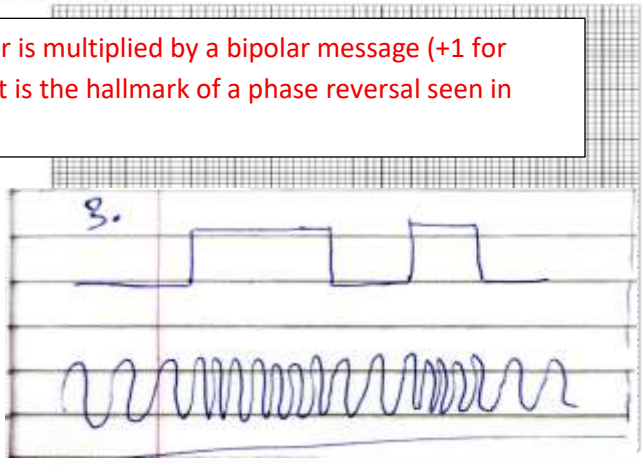
4. Compare the digital signal with the BPSK's signal envelope.
   If necessary, use the scope's Horizontal position control to view at least one transition of the digital signal.

What happens to the BPSK signal on the data stream's logic transitions?

The phase of the carrier signal shifts by **180**

What feature of the BPSK signal suggests that it's a DSBSC signal?

BPSK is functionally **DSBSC** because the carrier is multiplied by a bipolar message (+1 for Logic 1 and -1 for Logic 0). The 180 phase shift is the hallmark of a phase reversal seen in DSBSC when the message crosses zero.



Part B - Demodulating a BPSK signal using a product detector

As BPSK is really just DSB-SC (with a digital message instead of speech or music), it can be recovered using any of the DSBSC demodulation schemes.

1. Locate the Tuneable LPF module and turn its Cut-off Frequency Adjust control fully clockwise.
2. Set the Tuneable LPF module's Gain control to about the middle of its travel.
3. Modify the set-up as shown in Figure 12 below.

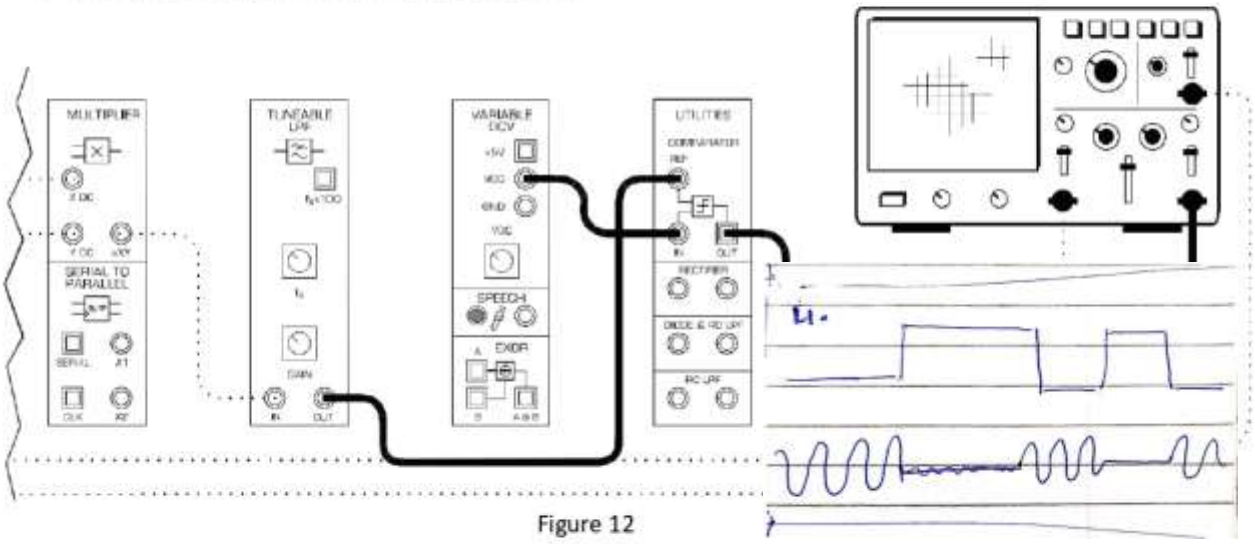


Figure 12

4. Set the Variable VDC module's Variable DC control to about the middle of its travel.
5. Compare the original and recovered signals. If they're not the same vary the Variable VDC module's Variable DC control until they are (ignoring phase shift).

Why is the recovered digital signal before comparator not a perfect copy of the original?

Why does varying the DC voltage on the comparator's input change the shape of the digital signal?

The recovery process involves a **Multiplier** and a **Tuneable LPF**. The filter is necessary to remove the high-frequency components of the multiplication, but it also slows down the transitions of the digital signal, causing rounding and phase delay.

Changing the **Reference DC voltage** moves the "decision threshold". Since the recovered signal has sloped (rounded) edges, a higher threshold will delay the "OFF to ON" transition and speed up the "ON to OFF" transition, effectively changing the **duty cycle** of the

It's possible to model noise being added to a signal in the channel of a communications system

6. Connect the set-up shown in Figure 13 but don't disconnect any of your existing wiring.
7. Unplug the patch lead to the output of the Multiplier module on the upper-half of the trainer and connect the noisy channel's input to it.
8. Unplug the patch lead to the output of the Multiplier module in the lower-half of the trainer and connect the noisy channel's output to it.
9. Compare the original and recovered data. If they're not the same, vary the Variable VDC module's Variable DC control until they are.
10. Unplug the scope's Channel-2 input from the comparator's output and connect it to the Adder module's output to observe the noisy BPSK signal.
11. increase the noise in the channel, observe the effect that this has on the BPSK signal and the recovered signal.

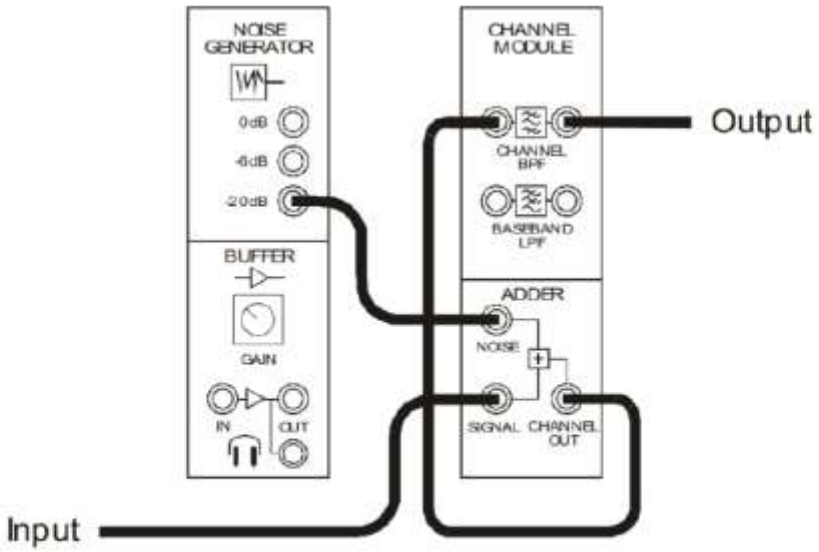
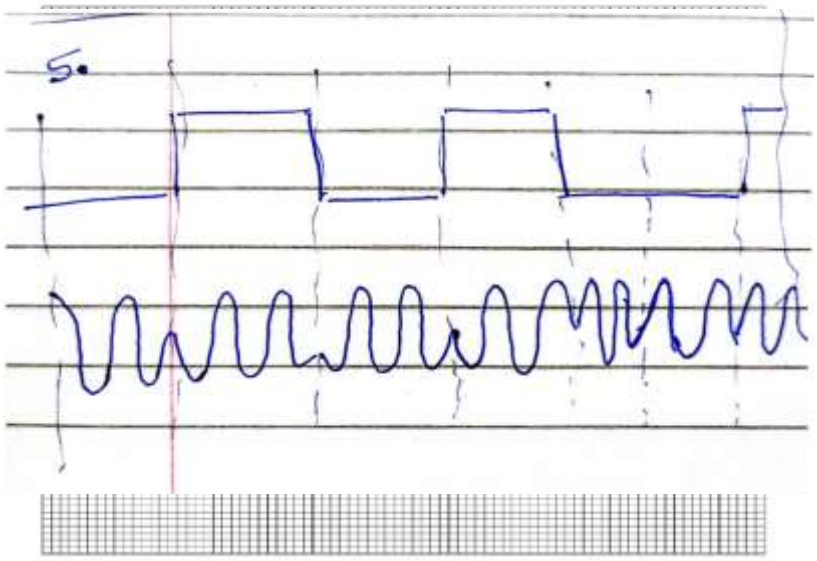


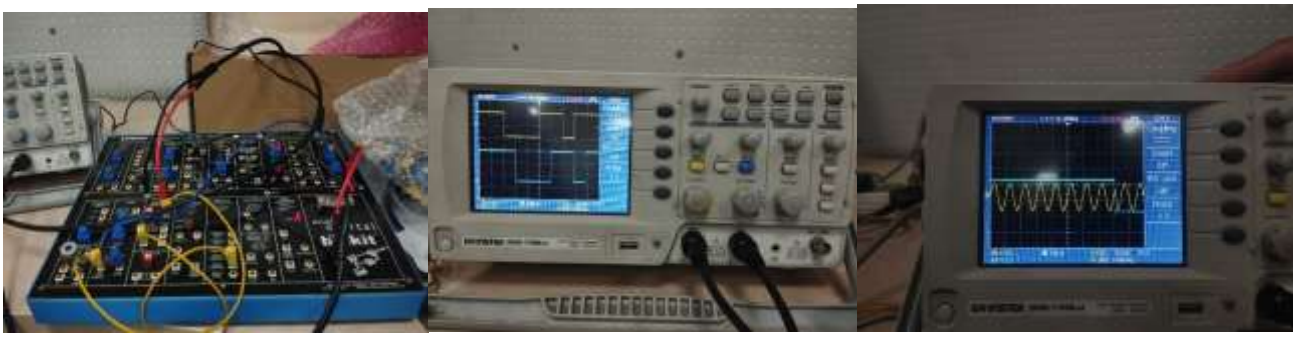
Figure 13



Pictures of output waveforms on oscilloscope





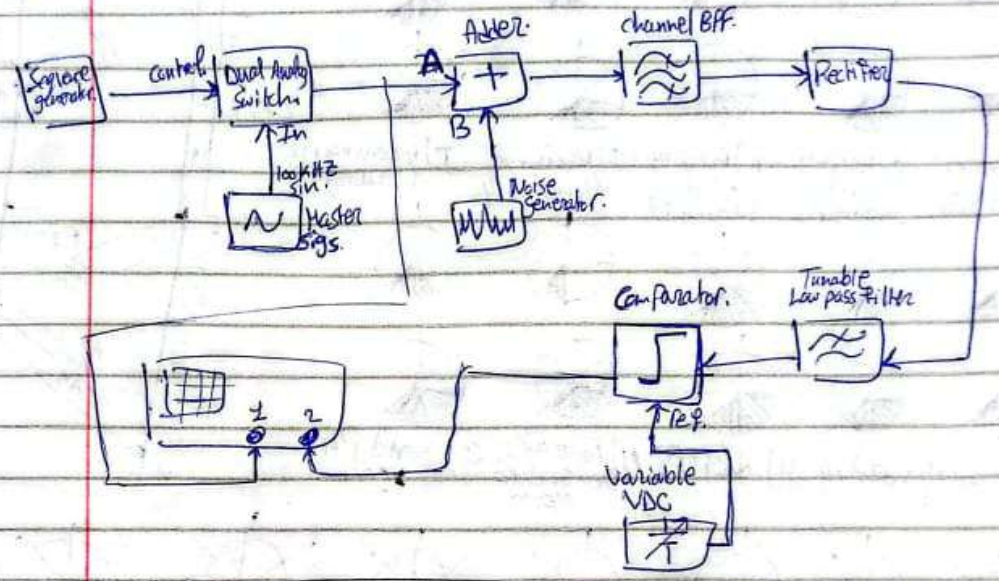




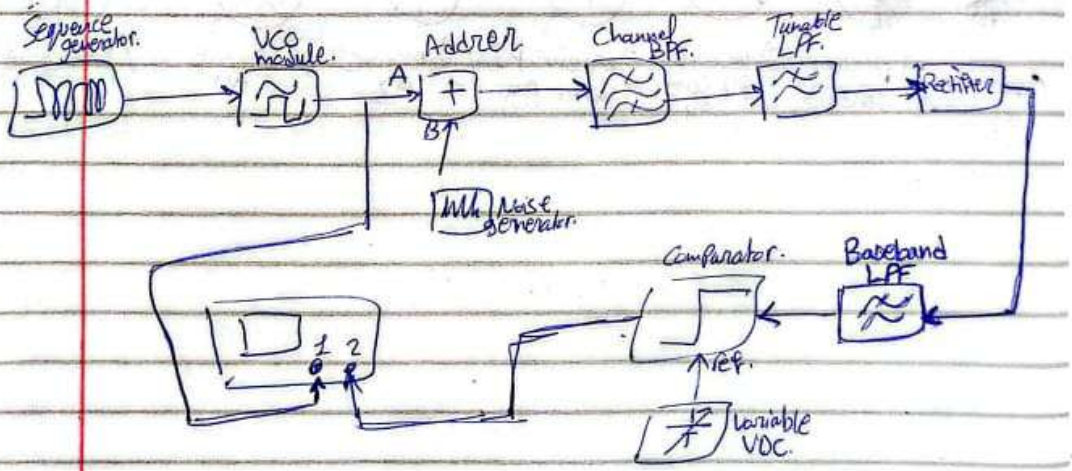
ADC report 4

Block Diagrams.

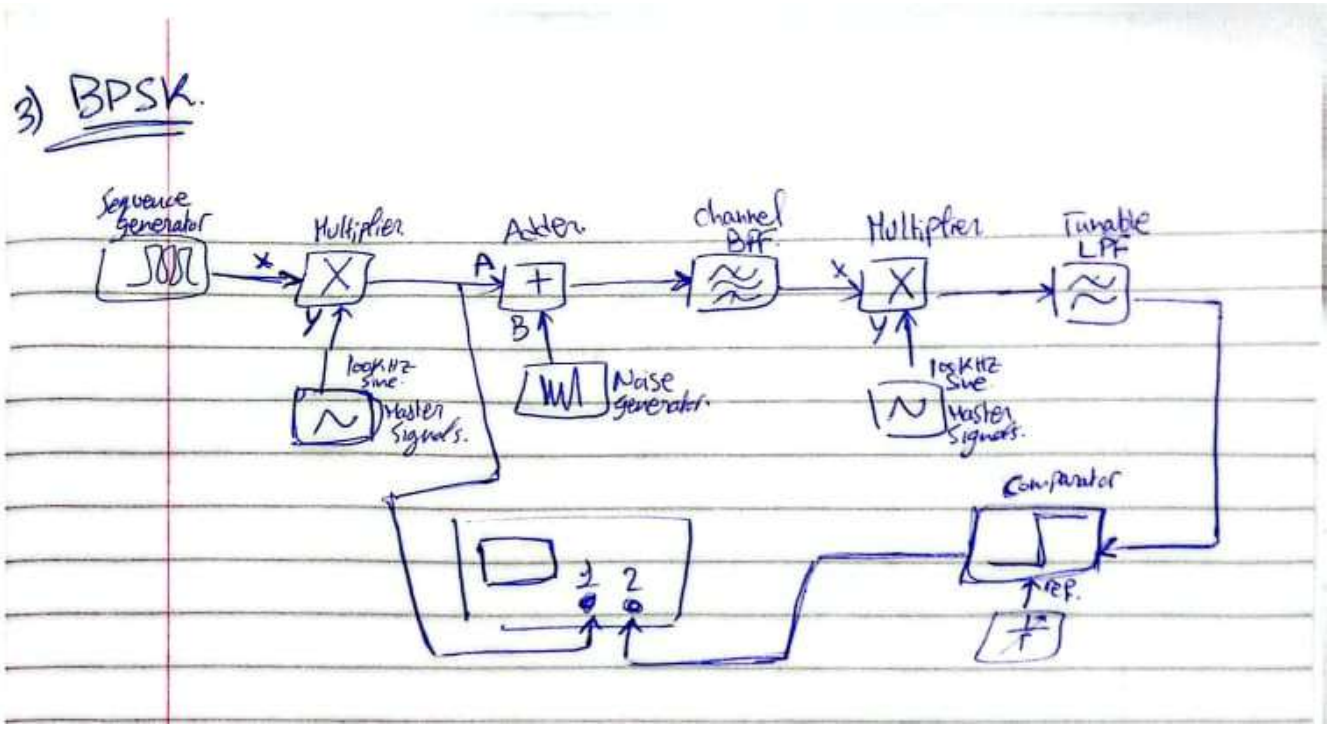
1) BASK.



2) BFSK



3) BPSK.


Sequence Generator
Multiplier
Adder
channel Aff.
Multiplier
Tunable LPF
x
y
A
B
100 KHz Sine
Master Signals.
Noise generator.
Comparator
1
2
Ref.